

Experiment 02: Series and Parallel Resistors

Amal Madhu
M.Tech VLSI Design

19 August 2026

Abstract

This experiment verifies the equivalent-resistance rules for series and parallel resistor networks using LTspice operating-point analysis. The series network contains a 9 V source and 1 k Ω and 2 k Ω resistors. The parallel network uses the same source voltage and resistor values in separate branches. Analytical calculations predict a 3 mA series current, voltage drops of 3 V and 6 V, and parallel branch currents of 9 mA and 4.5 mA. LTspice agrees with these values to numerical solver precision. The experiment demonstrates voltage division, current division, Kirchhoff's voltage law, and power distribution.

1 Introduction

The topology of a resistor network determines how voltage and current are distributed. In a series network, the same current flows through each element and the total resistance is the sum of the individual resistances. In a parallel network, each branch shares the same voltage and the total current is the sum of the branch currents. These rules are foundational for bias networks, sensor interfaces, current mirrors, feedback paths, and power-distribution networks.

The objective is to verify these relationships using a reproducible LTspice operating-point simulation and to compare the numerical results against hand calculations.

2 Theoretical Background

For two series resistors,

$$R_{eq,s} = R_1 + R_2. \quad (1)$$

For two parallel resistors,

$$R_{eq,p} = \frac{R_3 R_4}{R_3 + R_4}. \quad (2)$$

In the series network, the current is

$$I_s = \frac{V_1}{R_1 + R_2}, \quad (3)$$

and the individual voltage drops are $V_{R1} = I_s R_1$ and $V_{R2} = I_s R_2$. In the parallel network, the branch currents are $I_{R3} = V_2 / R_3$ and $I_{R4} = V_2 / R_4$, while the source current is their sum.

3 Circuit Under Test

The schematic contains two independent networks sharing only the ground reference. The series network uses $V_1 = 9\text{ V}$, $R_1 = 1\text{ k}\Omega$, and $R_2 = 2\text{ k}\Omega$. The parallel network uses $V_2 = 9\text{ V}$, $R_3 = 1\text{ k}\Omega$, and $R_4 = 2\text{ k}\Omega$.

The LTspice schematic and netlist are stored at:

```
simulation/series_parallel_resistors.asc  
simulation/series_parallel_resistors.net
```

4 Analytical Derivation

For the series network,

$$R_{eq,s} = 3\text{ k}\Omega, \quad I_s = \frac{9\text{ V}}{3\text{ k}\Omega} = 3\text{ mA}. \quad (4)$$

Thus,

$$V_{R1} = 3\text{ mA} \times 1\text{ k}\Omega = 3\text{ V}, \quad (5)$$

$$V_{R2} = 3\text{ mA} \times 2\text{ k}\Omega = 6\text{ V}. \quad (6)$$

For the parallel network,

$$R_{eq,p} = \frac{(1\text{ k}\Omega)(2\text{ k}\Omega)}{3\text{ k}\Omega} = 666.6667\text{ }\Omega. \quad (7)$$

The branch currents are

$$I_{R3} = \frac{9\text{ V}}{1\text{ k}\Omega} = 9\text{ mA}, \quad I_{R4} = \frac{9\text{ V}}{2\text{ k}\Omega} = 4.5\text{ mA}, \quad (8)$$

so the total parallel current is 13.5 mA. The expected series and parallel powers are 27 mW and 121.5 mW, respectively.

5 Simulation Methodology

An operating-point analysis is sufficient because the question concerns static DC relationships. The measurement directives are:

```
.op
.meas OP I_SERIES FIND I(R1)
.meas OP V_DROP_R1 FIND V(n_s0,n_s1)
.meas OP V_DROP_R2 FIND V(n_s1)
.meas OP I_PARALLEL_R3 FIND I(R3)
.meas OP I_PARALLEL_R4 FIND I(R4)
.meas OP V_PARALLEL FIND V(n_p)
```

No transient, AC, noise, temperature, or Monte Carlo analysis is needed for this ideal-network experiment.

6 Results

The LTspice operating-point log reports:

```
I(R1) = 0.0030000000002608 A
V(n_s0,n_s1) = 3 V
V(n_s1) = 6 V
I(R3) = 0.0089999999961257 A
I(R4) = 0.0044999999980628 A
V(n_p) = 9 V
```

The evidence images are stored under `assets/`.

7 Analytical Versus SPICE Comparison

The small differences in current are consistent with numerical solver precision. There is no physical-model discrepancy because all resistors are ideal.

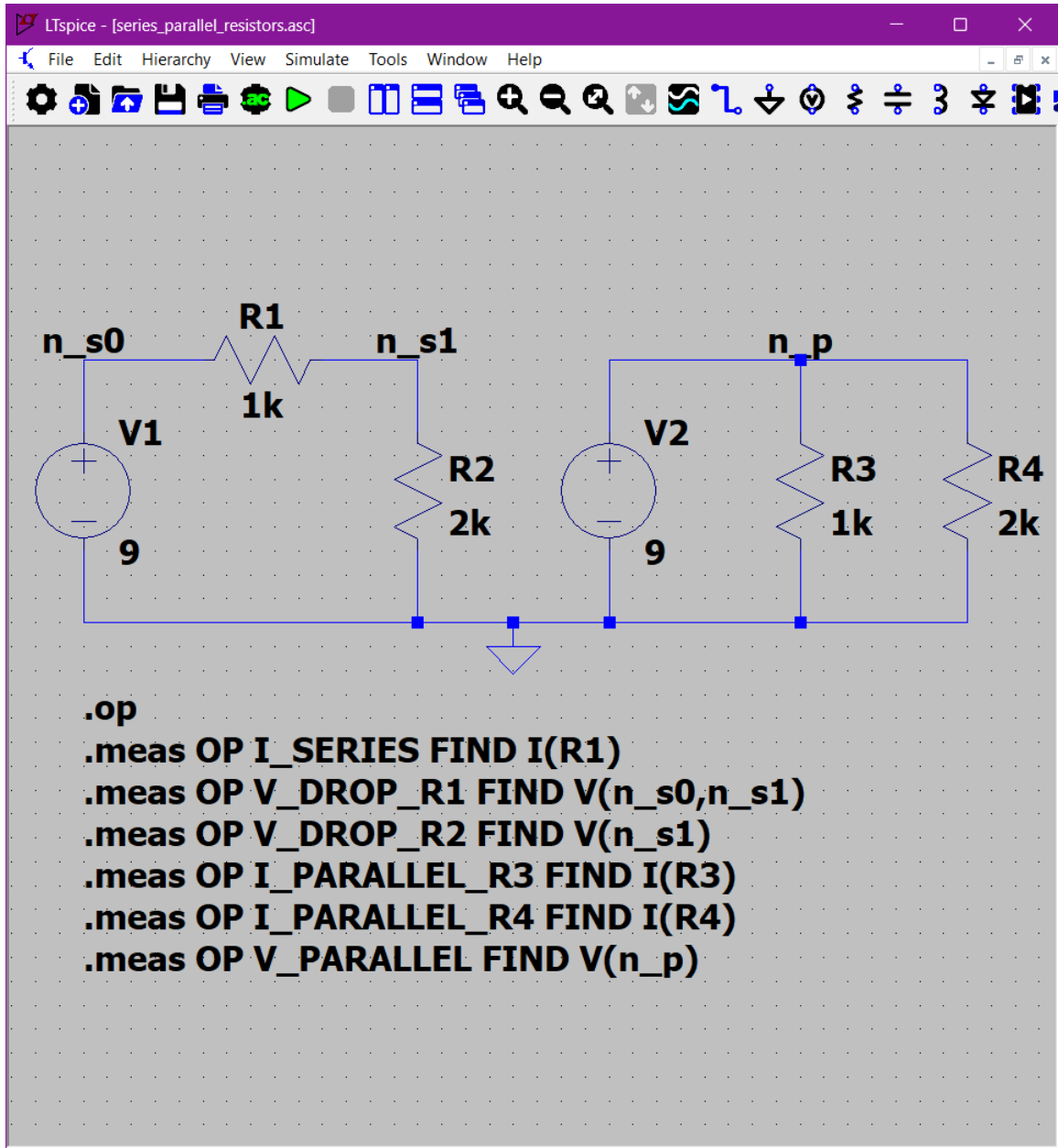


Figure 1: Series and parallel resistor networks used for the operating-point analysis.

Table 1: Comparison of analytical and LTspice results.

Quantity	Analytical	LTspice	Interpretation
Series current	3.000 000 mA	3.000 000 002 608 mA	Numerical agreement
R_1 voltage drop	3.000 000 V	3.000 000 V	Voltage division verified
R_2 voltage drop	6.000 000 V	6.000 000 V	KVL verified
R_3 branch current	9.000 000 mA	8.999 999 961 257 mA	Current division verified
R_4 branch current	4.500 000 mA	4.499 999 980 628 mA	Current division verified
Parallel voltage	9.000 000 V	9.000 000 V	Equal branch voltage verified

--- Operating Point ---		
V(n_p) :	9	voltage
V(n_s0) :	9	voltage
V(n_s1) :	6	voltage
I(R1) :	0.003	device_current
I(R2) :	0.003	device_current
I(R3) :	0.009	device_current
I(R4) :	0.0045	device_current
I(V1) :	-0.003	device_current
I(V2) :	-0.0135	device_current

Figure 2: Operating-point voltages and branch currents reported by LTspice.

```

LTspice 24.1.10 for Windows
Circuit: C:\Users\AMAL MADHU\Github_Current_Proj\01_Basic_Electronics\01_Resistor\02_Series_Parallel_Resistors\simulation\series_parallel_resistors.net
Start Time: Wed Aug 19 10:48:13 2026
solver = Normal
Maximum thread count: 12
tnom = 27
temp = 27
method = trap
Direct Newton iteration for .op point succeeded.
Total elapsed time: 0.136 seconds.

Files loaded:
C:\Users\AMAL MADHU\Github_Current_Proj\01_Basic_Electronics\01_Resistor\02_Series_Parallel_Resistors\simulation\series_parallel_resistors.net

i_series: I(R1)=0.00300000002608 at 9
v_drop_r1: V(n_s0,n_s1)=3 at 9
v_drop_r2: V(n_s1)=6 at 9
i_parallel_r3: I(R3)=0.008999999961257 at 9
i_parallel_r4: I(R4)=0.004499999980628 at 9
v_parallel: V(n_p)=9 at 9

```

Figure 3: Numerical measurement results from the LTspice output log.

8 Discussion

The simulation confirms that series resistors share current and divide voltage in proportion to resistance. It also confirms that parallel resistors share voltage and divide current inversely according to resistance. The two networks provide complementary tests of topology and conservation laws: KVL governs the voltage sum in the series path, while KCL governs the current sum in the parallel network.

9 Engineering Insights

These relationships are directly useful in mixed-signal IC design and hardware systems. Voltage-divider networks establish bias points, parallel devices distribute current, and equivalent-resistance reasoning simplifies larger networks before transistor-level analysis. The same discipline will later support MOSFET biasing, current mirrors, differential pairs, sensor front ends, feedback networks, and embedded power interfaces.

10 Limitations

The experiment uses ideal resistors and ideal voltage sources. It does not include source resistance, tolerance, temperature dependence, parasitic impedance, contact resistance, or measurement loading. Those effects should be introduced only when they become relevant to the engineering question.

11 Conclusion

The LTspice operating-point simulation verifies the analytical series and parallel resistor relationships. The measured series current, voltage drops, parallel branch currents, and parallel node voltage agree with hand calculations to numerical precision. The experiment establishes a reproducible network-analysis baseline for later circuit and mixed-signal studies.

References

- [1] Analog Devices, *Ltspice simulator*, Official simulator download and learning resources; no DOI assigned, Analog Devices, Inc., 2026. Accessed: Aug. 19, 2026. [Online]. Available: <https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
- [2] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*, 7th ed. Oxford University Press, 2014, Textbook reference; DOI not required for this book citation, ISBN: 9780199339136.